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Research paper

A trustworthy and explainable deep learning framework for skin lesion detection in smart dermatology

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ABSTRACT

The rapid evolution of artificial intelligence (AI) and deep learning profoundly impacts medical imaging, where it significantly enhances diagnostic accuracy. However, the effective deployment of AI systems in clinical settings, especially for skin lesion detection and diagnosis, requires not only high accuracy but also transparency and robustness to gain the trust of healthcare professionals. This is particularly crucial considering the challenges posed by varying sensor quality, lighting conditions, and lesion diversity. In this paper, we introduce a novel framework based on the You Only Look Once (YOLO) model that addresses these critical needs by enhancing both the explainability and performance of skin lesion detection models. Early and accurate identification of skin lesions is essential for the timely treatment and management of dermatological conditions. Traditional diagnostic methods, such as visual assessments by dermatologists, are often labor-intensive, subject to interpretative variability, and prone to inaccuracies, especially in cases involving atypical or subtle lesions. Our approach incorporates advanced data augmentation techniques to improve the model's generalization capabilities across diverse clinical conditions. Additionally, we integrate saliency maps to provide visual explanations of the model's predictions, allowing clinicians to understand the decision-making process and ensuring alignment with established clinical knowledge. Comparative analyses with the state-of-the-art models highlight the superior performance of our proposed framework, with significant improvements in the harmonic mean of precision and recall (F1-Score), and the Mean Average Precision (mAP50). The results underscore the effectiveness of our framework and how it advances the application of trustworthy AI in dermatology, paving the way for more reliable and informed clinical decisions in the diagnosis and treatment of skin conditions.

1. Introduction

The rapid advancement of deep learning has significantly transformed the field of artificial intelligence (AI), leading to groundbreaking developments in medical imaging and diagnostic practices. Deep learning models have proven to be highly effective in automating and enhancing the accuracy of medical image analysis, which is crucial for diagnosing a wide range of diseases. These models have been extensively applied across various medical domains, often surpassing traditional diagnostic methods in terms of speed, accuracy, and reliability.

In radiology, for example, deep convolutional neural networks (CNNs) have been employed to detect lung diseases from chest X-rays. The CheXNet model, developed by Rajpurkar et al. (2017), is a notable example, where a 121-layer CNN was trained to detect pneumonia, achieving performance comparable to that of expert radiologists. This model has since been adapted to identify multiple thoracic diseases, highlighting the versatility and power of deep learning in radiology. Similarly, in ophthalmology, deep learning has made significant strides, particularly in the screening and diagnosis of diabetic retinopathy. Gulshan et al. (2016) developed a deep learning algorithm that analyzed retinal fundus photographs, achieving sensitivity and specificity comparable to human experts. This system represents a critical advancement in automated eye disease screening, with potential implications for reducing the global burden of preventable blindness. Cardiology has also benefited from deep learning applications, especially in the analysis of echocardiograms. Madani et al. (2018) introduced a deep learning model capable of classifying echocardiogram views, which is essential for accurate interpretation and diagnosis of cardiac conditions. This work has laid the foundation for more advanced applications in automated cardiac imaging, demonstrating the potential of AI to enhance diagnostic accuracy and efficiency in cardiology.

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Dermatology, too, has seen substantial progress with the application of deep learning, particularly in the detection and classification of skin cancer. Esteva et al. (2017) conducted a landmark study where they trained a deep neural network on a large dataset of dermoscopic images to classify skin cancer with accuracy comparable to dermatologists. This model demonstrated the potential of deep learning to assist in early detection and improve patient outcomes by accurately identifying melanoma and other skin cancers. However, despite these advances, broader applications of deep learning in dermatology—particularly for skin diseases beyond cancer—remain underexplored. While skin cancer detection has garnered significant attention, other dermatological conditions such as acne, psoriasis, and dermatitis have received comparatively less focus.

Applying deep learning to the detection of skin lesions presents several technical challenges. One of the primary challenges is the wide variability in lesion appearance, which can range from subtle differences in color and texture to significant variations in size and shape. This variability makes it difficult for models to generalize across lesion types, particularly when trained on datasets with limited diversity. Furthermore, class imbalance is a prevalent issue, as conditions like acne may be overrepresented in datasets, while rarer conditions such as psoriasis or dermatitis are underrepresented. This can bias models towards common classes and degrade performance on clinically important but infrequent conditions. Image quality is another critical factor. Medical images are often captured under non-standardized conditions, leading to inconsistencies in lighting, resolution, and the presence of artifacts such as hair or shadows. These factors can obscure lesion features and reduce detection reliability. For deployment in real-world clinical workflows—including telemedicine and mobile health applications—models must be robust to a wide spectrum of imaging quality and modalities.

To address these challenges, we propose a novel You Only Look Once version 8 (YOLOv8)-based framework that enhances both the explainability and detection performance of skin lesion models. YOLOv8 was chosen due to its advanced feature pyramid network, which enables efficient detection across multiple scales—making it particularly suitable for the diverse visual characteristics of skin lesions. In addition, YOLOv8 features a lightweight architecture, anchor-free detection, and decoupled classification/localization heads, which improve both speed and flexibility compared to previous versions such as YOLOv7. These enhancements make it ideal for real-time diagnostic applications.

Our approach integrates several components to overcome the key challenges in skin lesion detection. First, we apply data augmentation techniques—including brightness adjustment, rotation, and noise addition—to simulate the variability found in real clinical environments and improve generalization. Second, to enhance interpretability, we incorporate saliency maps into the detection pipeline. These maps highlight the regions of an image that influenced the model's decision, providing clinicians with visual explanations of predictions and building trust in the system. Finally, to mitigate class imbalance, we incorporate focal loss during training, which emphasizes harder-to-classify examples and promotes balanced detection performance across rare and common lesion types.

The proposed framework was implemented and evaluated using a diverse dermatological image dataset, comprising both common and rare skin conditions. The dataset was curated to mitigate class imbalance, and extensive data augmentation was applied to simulate variability in clinical imaging conditions. The proposed YOLOv8-based model was benchmarked against a comprehensive set of state-of-the-art alternatives, including YOLOv9t, YOLOv10n-based real-time detector, and non-YOLO models such as RetinaNet, Faster R-CNN, and Vision Transformer. Evaluation results demonstrate that the proposed model consistently achieves superior performance across key detection metrics—F1-score, mean Average Precision at 50% IoU (mAP@50), and a well-balanced trade-off between precision and recall. Beyond accuracy, the model exhibits the lowest inference latency (2.5 ms per image) and shortest training time (0.35 h) among all compared models, confirming its suitability for real-time clinical applications. Additionally, explainability is supported through saliency map visualizations, which offer interpretability by highlighting clinically relevant regions that influenced predictions. A cross-dataset evaluation further demonstrates the model's generalization capability across diverse imaging conditions and sources. Collectively, these results validate the framework's potential as a robust, interpretable, and deployable solution for automated skin lesion detection in smart dermatology environments.

2. Related work

2.1. Computer vision models for clinical imaging/diagnosis

Computer vision has become a pivotal technology in clinical imaging, enabling automated analysis that enhances both the accuracy and efficiency of diagnostic processes. Traditionally, Convolutional Neural Networks (CNNs) have been the backbone of this advancement, with models like R-CNN and its variants playing a significant role in tasks such as segmentation and classification of medical images. These models excel at extracting local features from images, making them particularly useful for detecting specific abnormalities like tumors or lesions in various medical imaging modalities, including CT scans, MRIs, and X-rays (Ronneberger et al., 2015). In radiology, models like CheXNet have demonstrated the capability to detect pneumonia from chest X-rays, achieving performance comparable to that of expert radiologists (Rajpurkar et al., 2017). Similarly, in ophthalmology, deep learning algorithms have been employed to diagnose diabetic retinopathy, significantly improving early detection and potentially reducing the global burden of preventable blindness (Gulshan et al., 2016). Furthermore, in cardiology, deep learning models have been developed for tasks such as echocardiogram view classification, facilitating more accurate diagnosis of cardiac conditions (Madani et al., 2018). These successes across various medical domains underscore the transformative potential of computer vision models in clinical settings, driving the push towards more efficient and accurate diagnostic tools. Recent advancements have seen the introduction of Vision Transformers (ViTs) (Mohan et al., 2025; Khan and Khan, 2023), which overcome some of the limitations of CNNs by capturing long-range dependencies and spatial correlations within medical images. This capability is particularly important for segmenting complex and interconnected structures in medical images, such as organ boundaries or widespread lesions. ViTs, often combined with CNNs in hybrid models, have demonstrated improved performance in handling the intricacies of medical imaging data, such as subtle tissue differences and complex anatomical structures (Salahuddin et al., 2023). Despite these advancements, challenges remain, including the need for large annotated datasets, the high computational cost of training, and the lack of model interpretability.

The proposed framework aims to address these challenges by integrating explainability features and optimizing performance, even with smaller datasets, thereby enhancing the clinical applicability of computer vision models in dermatology.

2.2. Skin cancer detection

Skin cancer detection has been one of the most prominent areas of research in medical imaging, driven by the need for early and accurate diagnosis of melanoma. Deep learning models, particularly CNNs, have been highly successful in this domain. Esteva et al. were among the first to demonstrate that CNNs could achieve dermatologist-level accuracy in classifying skin cancer from dermoscopic images, setting a benchmark for future research (Esteva et al., 2017). Other models like ResNet and Inception have further pushed the boundaries of accuracy and have been integrated into mobile applications, making these technologies more accessible for real-time clinical use (Azad et al., 2023). Yu et al. (2017) developed a deep CNN model for melanoma detection

using a fully convolutional residual network (FCRN) with 16 residual blocks for better segmentation. Their method combined SVM and softmax classifiers, attaining 85.5% accuracy for melanoma classification with segmentation and 82.8% without it. DeVries and Ramachandram (2017) proposed a multi-scale CNN model based on the Inception v3 network, which was fine-tuned on two scales: coarse for shape and context and finer for textural details of lesions. Mahbod et al. (2019) introduced a method to extract deep features from pre-trained CNNs like AlexNet, ResNet-18, and VGG16 for skin lesion classification. These features were used to train a multi-class SVM classifier, which was tested on the ISIC 2017 dataset, showing an area under the curve (AUC) of 97.55% for seborrheic keratosis (SK) and 83.83% for melanoma. Another system built on a pre-trained ResNet-152 model classified 12 types of skin lesions, achieving an AUC of 0.99 for conditions such as hemangioma, pyogenic granuloma (PG), and intraepithelial carcinoma (IC) (Mendes and da Silva, 2018). Dorj et al. (2018) presented a technique using a pre-trained AlexNet for feature extraction, followed by error-correcting SVM for classification. The system performed well on skin cancers such as SCC, actinic keratosis (AK), and BCC, with average sensitivity, specificity, and accuracy of 95.1%, 98.9%, and 94.17%, respectively. Kalouche (2016) proposed fine-tuning the VGG-16 model by adding three final layers and five convolutional blocks, which resulted in a 78% accuracy rate for melanoma detection. Another deep CNN system was trained to differentiate between normal skin and lesion images, achieving 86.67% accuracy. A detailed list of CNN-based skin cancer detection systems can be found in Hameed et al. (2016), Shah et al. (2023), Bhatt et al. (2023), Nazari and Garcia (2023), Behara et al. (2024), Verma et al. (2025), Zafar et al. (2023), Rezk et al. (2023), Hussain and Toscano (2024), Ghosh et al. (2024), Alzamili and Ruhaiyem (2025), Vayadande et al. (2024) and Kandhro et al. (2024).

However, while skin cancer detection has seen substantial research and development, the application of deep learning to other dermatological conditions remains relatively underexplored, highlighting an area ripe for further investigation.

2.3. Skin lesion detection

While much attention has been given to skin cancer detection, the broader field of skin lesion segmentation (SLS) and classification has seen recent developments. For instance, Akram et al. (2024) introduced a three-step framework, starting with multilevel contrast stretching for foreground and background separation, followed by a weighting criterion involving central distance and texture features, and concluding with advanced feature extraction. Similarly, Shan et al. (2020) proposed FC-DPN, a hybrid model combining fully convolutional and dual-path networks, achieving notable accuracy on the ISBI 2017 and PH2 datasets. In another approach, Ramya et al. (2021) used discrete wavelet decomposition to remove unwanted data, while Ramadan and Aly (2022) introduced DGCU-Net, a dual-path model that outperformed traditional U-Net models. Additionally, Yap et al. (2018) developed a late fusion model based on ResNet-50, showing improved performance in melanoma detection and multi-class classification compared to previous methods.

Further improvements in classification were seen with Nasiri et al. (2020), who introduced a 16-layer CNN to enhance malignancy detection, and Lima et al. (2021), who used squeeze-and-excite mechanisms within CNNs. Lima et al. (2021) tackled data augmentation using a GAN-based model, achieving significant sensitivity improvements. Talavera-Martínez et al. (2022) proposed a CNN-based approach to classify skin lesions into asymmetry categories, while Zhuang et al. (2022) advanced the field with a cost-sensitive multi-classifier fusion framework, yielding superior results on the ISIC 2019 dataset.

Hasan et al. (2022) presented Dermo-Expert, a hybrid CNN model with high AUC scores, while Khan et al. (2022) improved lesion contrast using color segmentation, achieving strong classification results. Gottumukkala et al. (2022) introduced BLSNet, which achieved high accuracy on the ISIC 2019 and PH2 datasets. Similarly, Tang et al. (2022) and Datta et al. (2021) developed models incorporating patient metadata and attention mechanisms, respectively, both yielding improved accuracy. Ding et al. (2021) introduced deep attention networks with loss weighting strategies to address class imbalance, while Wang et al. (2022) developed a multi-modal fusion model that further improved AUC scores. Wei et al. (2022) introduced a dual-attention model focused on local features, and Nakai et al. (2022) proposed a deep bottleneck transformer, achieving high accuracy on the ISIC and HAM10000 datasets. Additionally, Khouloud et al. (2022) combined attention W-Net and Inception-ResNet blocks, while Qin et al. (2020) employed GAN-based data augmentation to enhance performance.

Several other studies, including those by Serte and Demirel (2020), Rodrigues et al. (2020), and Jasil and Ulagamuthalvi (2021), explored various deep learning models and transfer learning techniques, achieving competitive results across multiple datasets. Notably, Khan et al. (2021) introduced a two-module system for segmentation and classification, and Varma et al. (2022) developed SLDCNet, achieving outstanding accuracy and specificity. Finally, Benyahia et al. (2022) and Al-saedi and Savaş (2022) employed ensemble learning and deep learning comparisons, respectively, with DenseNet121 achieving the highest accuracy. While these studies reflect significant advancements in SLS, they did not consider the classification of the recognition of the diseases considered in this study and dataset. The proposed YOLOv8-based framework improves the detection and classification of these skin lesions in the presence of class imbalance, data scarcity, etc.

2.4. Explainability in clinical imaging/diagnosis

Explainability is a critical aspect of deploying AI in clinical settings, where the "black-box" nature of deep learning models can be a significant barrier to adoption. Techniques such as saliency maps, Grad-CAM (Gradient-weighted Class Activation Mapping), and attention mechanisms have been developed to provide visual explanations of model decisions, highlighting the areas of an image that most influenced the prediction (Salahuddin et al., 2023). These techniques are particularly important in clinical settings. The need for transparency in AI-driven decision-making processes is paramount, as healthcare professionals must understand and trust the model's outputs to make informed decisions. In clinical imaging, explainability techniques such as saliency maps have been widely adopted to provide insights into which parts of the image most influenced the model's predictions (Samek et al., 2017). These visual tools are particularly valuable in dermatology, where the visual characteristics of skin lesions—such as texture, shape, and color—are crucial for diagnosis. By highlighting these features, saliency maps help bridge the gap between the model's decision-making process and clinical reasoning, thereby enhancing the trustworthiness of AI systems.

The integration of explainability into the proposed framework not only improves the transparency of the model but also ensures that its predictions align with established clinical knowledge, making it a more reliable tool for dermatological diagnosis.

3. Background

You Only Look Once (YOLO) is a pioneering real-time object detection system that reframes object detection as a single regression problem, offering an alternative to traditional sliding window-based approaches and region proposal-based networks like R-CNN. YOLO processes the entire image in a single forward pass of the neural network, directly predicting the bounding boxes and class probabilities. This approach contrasts with previous methods that required multiple stages, such as proposing regions and then classifying them, leading to slower performance. YOLO's architecture is fundamentally built on a convolutional neural network (CNN) backbone, typically derived from

classification networks like Darknet, designed to handle both the spatial and channel dimensions effectively.

At the core of YOLO's architecture is the division of the input image into an $S \times S$ grid. Each grid cell is responsible for predicting B bounding boxes and their associated confidence scores, where each bounding box is parameterized by four coordinates (x, y, w, h) and a confidence score C. The coordinates (x, y) represent the center of the bounding box relative to the grid cell, while w and h are the width and height of the bounding box normalized by the image dimensions. The confidence score C is defined as the product of the probability that an object is present in the bounding box and the Intersection over Union (IoU) between the predicted box and the ground truth.

Each grid cell also predicts conditional class probabilities $P(\text{Class}_i|\text{Object})$ for each of the C classes in the dataset. The output of the network is a tensor of dimensions $S \times S \times (B \times 5 + C)$, where the first $B \times 5$ components correspond to the bounding box predictions, and the remaining C components represent the class probabilities. This formulation allows YOLO to predict multiple objects in the same image with a fixed, consistent architecture.

YOLO's key advantage lies in its speed, as it can process images in real-time, making it suitable for applications requiring low-latency object detection, such as autonomous driving and video surveillance. This speed is achieved because YOLO processes the entire image at once, instead of focusing on multiple regions or scales, which allows it to capture global context and reduce false positives. However, this design also introduces some challenges. YOLO tends to struggle with small objects, particularly those that fall between grid cells, as the fixed grid size can limit its ability to accurately localize these objects. Additionally, while YOLO is less prone to background errors, it may produce more localization errors compared to region proposal-based methods.

Over time, the YOLO framework has evolved to address some of its limitations and to improve both speed and accuracy. YOLOv2 (YOLO9000) introduced anchor boxes and batch normalization, improving its ability to detect objects at different scales. YOLOv3 further enhanced the architecture by using a deeper network and multi-scale predictions, allowing it to detect smaller objects more accurately. YOLOv4 and YOLOv5 introduced additional optimizations, such as better backbone networks, optimized training processes, and more advanced post-processing techniques like Mosaic data augmentation. The latest versions, including YOLOv8, continue to refine the balance between detection speed and accuracy, making YOLO a robust choice for real-time object detection tasks.

In conclusion, YOLO's innovative approach to object detection has made it a cornerstone in the field, particularly for applications requiring fast, real-time processing. Despite some trade-offs in localization accuracy, its ability to perform end-to-end object detection in a single pass offers advantages over traditional methods.

4. Dataset

In our study, we leveraged a publicly available and diverse dataset of skin lesion images, obtained from various medical databases and research repositories (Gk, 2024). This dataset covers different skin diseases, as outlined in Table 1. Each image is meticulously annotated with the corresponding lesion class, ensuring high-quality data for model training. The dataset comprises a total of 1101 images across six distinct classes: Acne, Wart, Leprosy, Psoriasis, Keratosis Pilaris, and Atopic Dermatitis. The class distribution is as follows: Acne is the most represented class with 1208 annotations, which constitutes approximately 38.6% of the entire dataset. Wart follows as the second most prevalent class with 807 annotations, accounting for about 25.8% of the dataset. Leprosy includes 643 annotations, representing around 20.5% of the dataset. Psoriasis, with 261 annotations, makes up 8.3% of the dataset and is labeled as underrepresented, indicating a need for attention during model training to avoid bias. Keratosis Pilaris comprises 137 annotations, which is about 4.37% of the dataset, and is similarly marked as underrepresented. Lastly, Atopic Dermatitis is the least represented class, with only 74 annotations, accounting for a mere 2.36% of the dataset.

The disparity in class representation highlights a potential challenge of class imbalance during model training. Classes with fewer samples, such as Psoriasis, Keratosis Pilaris, and Atopic Dermatitis, are at risk of being overshadowed by the more prevalent classes like Acne, Wart, and Leprosy. This could lead to the model developing a bias towards the more dominant classes, resulting in suboptimal performance for the underrepresented classes.

5. Methodology

The methodology employed in this research leverages the YOLOv8 model for detecting and classifying a variety of skin lesions, including Acne, Atopic Dermatitis, Keratosis Pilaris, Leprosy, Psoriasis, and Wart. This process is structured into two primary phases: Preprocessing and YOLOv8-based Detection. Additionally, the methodology addresses specific challenges associated with the dataset used, such as class imbalance, variation in image quality, and the complexity of skin lesion appearances.

5.1. Preprocessing

The preprocessing phase is critical to ensuring that the input images are well-prepared for the skin lesion detection model. The dataset comprises various skin lesion images captured under different conditions, leading to variability in lighting, resolution, image quality, and the presence of artifacts like hair. Addressing these challenges through careful preprocessing is essential to improving the model's accuracy (Hoshyar et al., 2014; Weiss et al., 2016). Additionally, balancing the augmentation techniques to enhance generalization without overfitting remains a key concern, particularly given the diverse nature of the dataset (Barata et al., 2015).

5.1.1. Image rescaling and normalization

To ensure consistency across the dataset, each image is resized to a standard dimension, $W \times H$, as required by the YOLOv8 model. This rescaling step is critical because the images in the Fredrick Skin Disease Dataset vary significantly in their original size and resolution. Resizing all images to a fixed size ensures that the model receives input in a uniform format, which is essential for both stable training and accurate inference. The rescaling process involves adjusting the spatial dimensions of the image while maintaining its aspect ratio to prevent distortion. This is achieved by padding the resized image with a border of a constant value (typically zero, corresponding to black) to fit the target dimensions $W \times H$. Formally, if an image I has original dimensions $W_o \times H_o$ and needs to be resized to $W \times H$, the scaling factor s is computed as:

$$s = \min\left(\frac{W}{W_o}, \frac{H}{H_o}\right)$$

The image is then resized to dimensions $W' = s \times W_o$ and $H' = s \times H_o$, and padded to the target size. This ensures that the entire image content is preserved without distortion, while also fitting the YOLOv8 input requirements (Matsunaga et al., 2017; Lucknavalai and Schulze, 2020) (see Fig. 1).

Following rescaling, pixel values of the image are normalized to the range [0, 1]. Normalization is a standard preprocessing step in convolutional neural networks (CNNs) that helps to stabilize and accelerate the training process. In the context of the YOLOv8 model, normalization adjusts the input image's pixel intensity values so that they fall within a common range, reducing the likelihood of numerical instability during training. This is particularly important for datasets like the Fredrick Skin Disease Dataset, where images might come from various sources with different lighting conditions and exposure levels.

Table 1
Description of the dataset.

Class	Image	Description
Acne		A prevalent skin disorder marked by the presence of pimples, blackheads, whiteheads, and cysts, typically found on the face, neck, chest, and back. Often linked to excess oil production, clogged pores, and bacterial infection.
Atopic Dermatitis		Also known as eczema, this chronic inflammatory condition results in dry, itchy, and inflamed patches of skin, often triggered by environmental factors or allergies. It frequently affects areas such as elbows, knees, and neck.
Keratosis Pilaris		A benign skin condition characterized by small, rough bumps typically found on the upper arms, thighs, buttocks, and cheeks. These bumps occur due to the buildup of keratin, a protein in the skin.
Leprosy		Also known as Hansen's disease, this chronic infectious disease is caused by the bacterium *Mycobacterium leprae*, primarily affecting the skin, nerves, and mucous membranes. If left untreated, it can lead to severe skin lesions, nerve damage, and deformities.
Psoriasis		A chronic autoimmune disorder that accelerates the growth of skin cells, leading to the formation of thick, red, scaly patches. It commonly affects the scalp, elbows, knees, and lower back, often accompanied by itching and inflammation.
Wart		Benign growths caused by the human papillomavirus (HPV) that can appear anywhere on the body. Warts are typically rough, raised, and vary in size, shape, and color. They are contagious and can spread through direct contact with infected surfaces or skin.

Formally, if I represents the original image with pixel values $p(i, j, k)$ where $i \in [1, H], j \in [1, W], k \in [1, 3]$ (corresponding to the three color channels: red, green, and blue), the normalized image I' is defined as:

$$p'(i, j, k) = \frac{p(i, j, k)}{255}$$

This normalization process transforms the pixel values from the original range of [0, 255] to the normalized range of [0, 1]. In cases where the dataset might include images with non-standard pixel ranges (such as 16-bit images), additional steps are taken to scale the pixel values accordingly before applying the normalization (Afza et al., 2022).

Moreover, normalization also helps in mitigating the effects of varying image brightness and contrast, which can otherwise lead to challenges in feature extraction during the convolutional operations of the YOLOv8 model. By ensuring that all input images share a common scale of pixel intensity, the model can focus on learning relevant features related to skin lesions without being influenced by irrelevant variations in image illumination (Codella et al., 2018; Hoshyar et al., 2014).

An additional step that could be incorporated to further enhance the model's robustness is mean subtraction and variance scaling, where the dataset's global mean μ and standard deviation σ of pixel values are computed across the entire training set. Each pixel value in the image is then standardized as follows:

$$p''(i, j, k) = \frac{p'(i, j, k) - \mu_k}{\sigma_k}$$

where μ_k and σ_k represent the mean and standard deviation for each color channel k. This step centers the data around zero and scales it to have a unit variance, further improving the convergence properties of the model during training (Lee et al., 1997; Hoshyar et al., 2014).

One of the challenges in the dataset is the presence of hair artifacts, which can obscure important details of skin lesions, leading to inaccurate detections. To address this, a hair removal technique is applied using morphological operations (Lee et al., 1997; Hoshyar et al., 2014). A binary hair mask I_H is generated by thresholding, followed by operations such as dilation and erosion to clean the image of hair. This process ensures that the model focuses on the lesions rather than being distracted by extraneous elements. The cleaned image I_{clean} is computed as:

$$I_{\text{clean}} = I - \mathcal{M}(I_H)$$

where $\mathcal{M}(I_H)$ represents the morphological operations applied (Hoshyar et al., 2014).

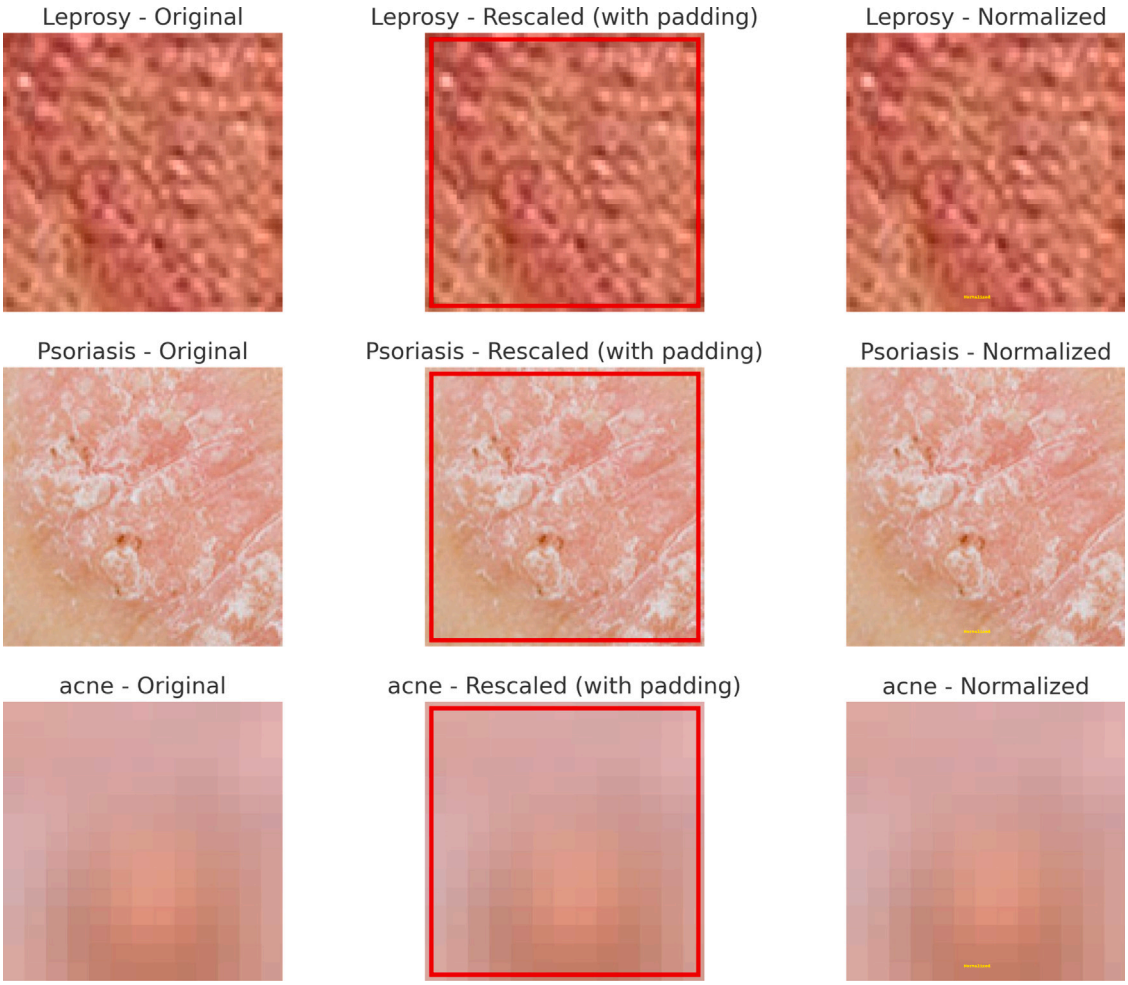


Fig. 1. Preprocessing steps for different classes of skin lesions. The figure illustrates the preprocessing steps applied to images from three different classes (Leprosy, Psoriasis, and Acne) within the dataset. The first column shows the original raw images. The second column demonstrates the rescaled images, where a red border highlights the padding added to maintain aspect ratio while fitting the target dimensions. The third column displays the normalized images, with a label indicating that pixel values have been adjusted to the [0, 1] range. These preprocessing steps are crucial for ensuring consistent input data for the YOLOv8 model, enhancing its ability to accurately detect and classify skin lesions.

5.1.2. Data augmentation

We employed a combination of geometric and photometric transformations to enhance the model's generalizability across diverse skin tones and lighting conditions. Specifically, augmentations such as rotation, random cropping, contrast adjustments, and Gaussian noise were applied to simulate real-world variations in medical imaging. Each augmentation technique was carefully tuned to ensure that the model could handle distortions while maintaining the diagnostic integrity of the lesions. For instance, contrast augmentation was particularly beneficial for improving the model's performance on underexposed images, which often occur in real-world clinical scenarios (Shen et al., 2021; Gonzalez et al., 2020).

Flipping the images horizontally and vertically is the first augmentation technique applied. This method is crucial because skin lesions can appear in various orientations, depending on how an image is captured. Mathematically, horizontal flipping is achieved by reversing the order of columns in the image matrix, while vertical flipping involves reversing the order of rows. For an image represented as a matrix I of size $H \times W$, where H is the height and W is the width, horizontal flipping is formalized as:

$$I'_h(i,j) = I(i, W - j + 1),$$

and vertical flipping as:

$$I'_v(i,j) = I(H - i + 1, j).$$

This augmentation helps the model learn orientation-invariant features, which is particularly important given the variability in how lesions may be positioned during image capture. The challenge addressed here is the inherent variability in lesion presentation, and the novelty of applying both horizontal and vertical flips lies in ensuring that the model is not biased towards a particular orientation, thus improving its ability to generalize across different perspectives (Perez and Wang, 2017).

Rotation is another augmentation technique that enhances the model's robustness to variations in image orientation. In this study, two types of rotation are applied: small random rotations within a specified range and 90° rotations in specific directions. Small random rotations are applied by rotating the image by an angle θ, which is uniformly sampled from the range $[-15°, +15°]$. The mathematical representation of this rotation involves transforming the coordinate grid of the image using a rotation matrix. For a point (x, y) in the image, the rotated coordinates (x', y') are computed as:

$$\begin{bmatrix} x' \\ y' \end{bmatrix} = \begin{bmatrix} \cos\theta & -\sin\theta \\ \sin\theta & \cos\theta \end{bmatrix} \begin{bmatrix} x \\ y \end{bmatrix}.$$

In addition to these small rotations, the images are also subjected to exact 90° rotations (clockwise, counter-clockwise, and upside down). The 90° rotations are implemented by rearranging the pixels according to specific rules: clockwise rotation by 90° is given by:

$$I'_{cw90}(i,j) = I(j, H - i + 1),$$

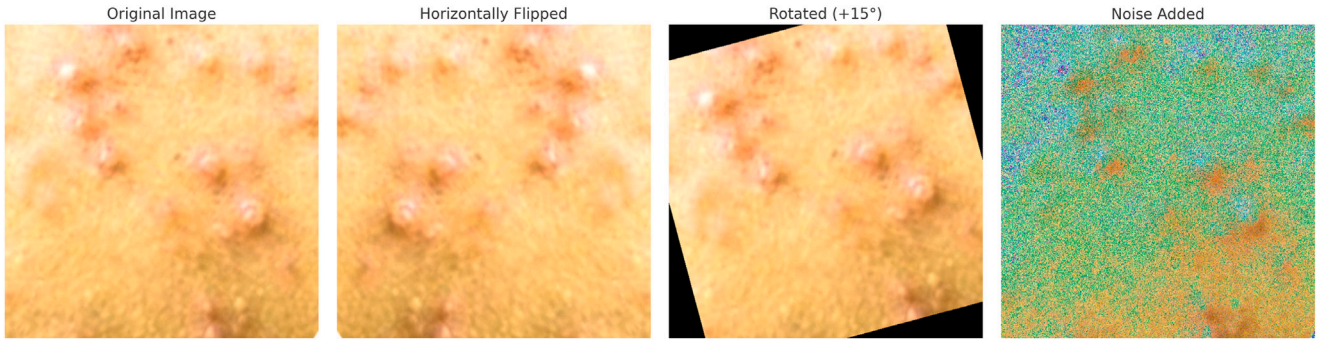


Fig. 2. Data augmentation. This figure shows the application of various data augmentation techniques to images representing different skin types or conditions. For each image, the original version is displayed alongside its augmented versions: horizontally flipped, rotated by 15 degrees, and with added noise. These augmentations help create a more diverse training dataset, enabling the model to generalize better across different skin types and imaging conditions.

counter-clockwise rotation by 90° is given by:

$$I'_{ccw90}(i,j) = I(W - j + 1, i),$$

and upside-down rotation by 180° is given by:

$$I'_{ud180}(i,j) = I(H - i + 1, W - j + 1).$$

The novelty of combining small random rotations with precise 90° rotations lies in their ability to simulate common camera orientation variations, which are particularly relevant in clinical settings where images may be taken from different angles. This dual approach to rotation ensures that the model is resilient to both slight and significant orientation changes, a challenge that is often encountered in real-world applications (Buslaev et al., 2020).

Brightness adjustment is another key augmentation technique used to simulate variations in lighting conditions. In this study, brightness is uniformly adjusted by a factor β, which is randomly sampled from the range $[-0.15, +0.15]$ for moderate adjustments and from $[-0.25, +0.25]$ for more extreme adjustments. The mathematical expression for brightness adjustment is:

$$I' = I + \beta,$$

where β is a scalar added to all pixel values in the image. By simulating both underexposed and overexposed conditions, this augmentation allows the model to learn how to handle images captured in suboptimal lighting environments, a common challenge in clinical practice. The novelty here lies in the use of two distinct ranges for brightness adjustment, which allows for the simulation of a broader spectrum of lighting conditions. This approach ensures that the model is not only robust to minor lighting variations but also capable of handling more significant deviations, thereby improving its performance in diverse clinical settings (Gonzalez et al., 2020).

Exposure adjustment, closely related to brightness adjustment, refers to the overall luminance of the image and is similarly adjusted by scaling the pixel values. In this study, exposure is adjusted by multiplying the image by a factor $1 + \alpha$, where α is sampled from the range $[-0.15, +0.15]$. The mathematical representation is:

$$I' = I \times (1 + \alpha).$$

This adjustment simulates different exposure levels, which might occur due to varying camera settings or environmental lighting conditions. The challenge addressed by this augmentation is the inconsistency in exposure that can significantly affect the visibility of skin lesions. The novelty of incorporating exposure adjustment lies in its ability to prepare the model for handling images with varying exposure levels, which is crucial for achieving reliable performance across different imaging devices and settings (Shen et al., 2021) (see Fig. 2).

Finally, noise addition is a data augmentation technique used to introduce random variations into the input images, simulating imperfections that might be present in real-world scenarios, such as those caused by imaging sensors or environmental conditions during image capture. This technique is particularly important in medical imaging because it helps the model become more robust to various types of noise that could obscure important features, such as skin lesions. Noise addition can be mathematically formalized by introducing a noise matrix N to the original image I. For each pixel $p(i,j,k)$ in the image I, where i and j represent the spatial coordinates, and k represents the color channel (e.g., red, green, blue), the noise-augmented pixel value $p'(i,j,k)$ is given by:

$$p'(i,j,k) = p(i,j,k) + N(i,j,k)$$

Here, $N(i,j,k)$ is a random variable representing the noise added to the pixel. This noise is typically sampled from a normal distribution:

$$N(i,j,k) \sim \mathcal{N}(0,\sigma^2)$$

where $\mathcal{N}(0,\sigma^2)$ denotes a Gaussian (normal) distribution with a mean of 0 and a variance of σ^2. The mean of 0 ensures that the noise does not introduce a systematic bias in the pixel values, while the variance σ^2 controls the intensity of the noise. A larger variance results in stronger noise, while a smaller variance leads to subtler noise (Perez and Wang, 2017).

To address the challenge of simulating realistic noise without compromising training data quality, the proposed approach uses partial noise addition. Noise is selectively applied to a subset of pixels, typically up to 0.1% of the total pixel count, to introduce subtle variations that mimic real-world scenarios like sensor noise or environmental factors without overwhelming the image. This targeted noise application strikes a balance between realism and practicality. By adding noise to only a small percentage of pixels, the model is exposed to imperfections it might encounter in real-world images while preserving the essential features needed for accurate lesion detection. Excessive noise could obscure critical features, making learning difficult, while insufficient noise might not provide enough variability to enhance robustness. This method ensures the model experiences realistic imperfections, improving its ability to generalize and perform reliably under various imaging conditions (Buslaev et al., 2020; Gonzalez et al., 2020).

By training with noisy images, the model learns to focus on relevant features, even in the presence of distractions or artifacts. This is crucial in medical imaging, where noise from patient movement, poor lighting, or hardware limitations can affect image quality. The approach helps the model disregard irrelevant variations and focus on consistent characteristics of skin lesions. The considered data augmentation techniques are meticulously designed to address the specific challenges associated with medical imaging, particularly in the context of skin lesion detection. The combination of these techniques adopted in this study creates a more resilient model capable of performing reliably in diverse clinical environments (Shen et al., 2021; Gonzalez et al., 2020). These augmentations effectively tripled the size of the original dataset, thereby reducing the risk of overfitting and improving the model's ability to generalize to unseen data.

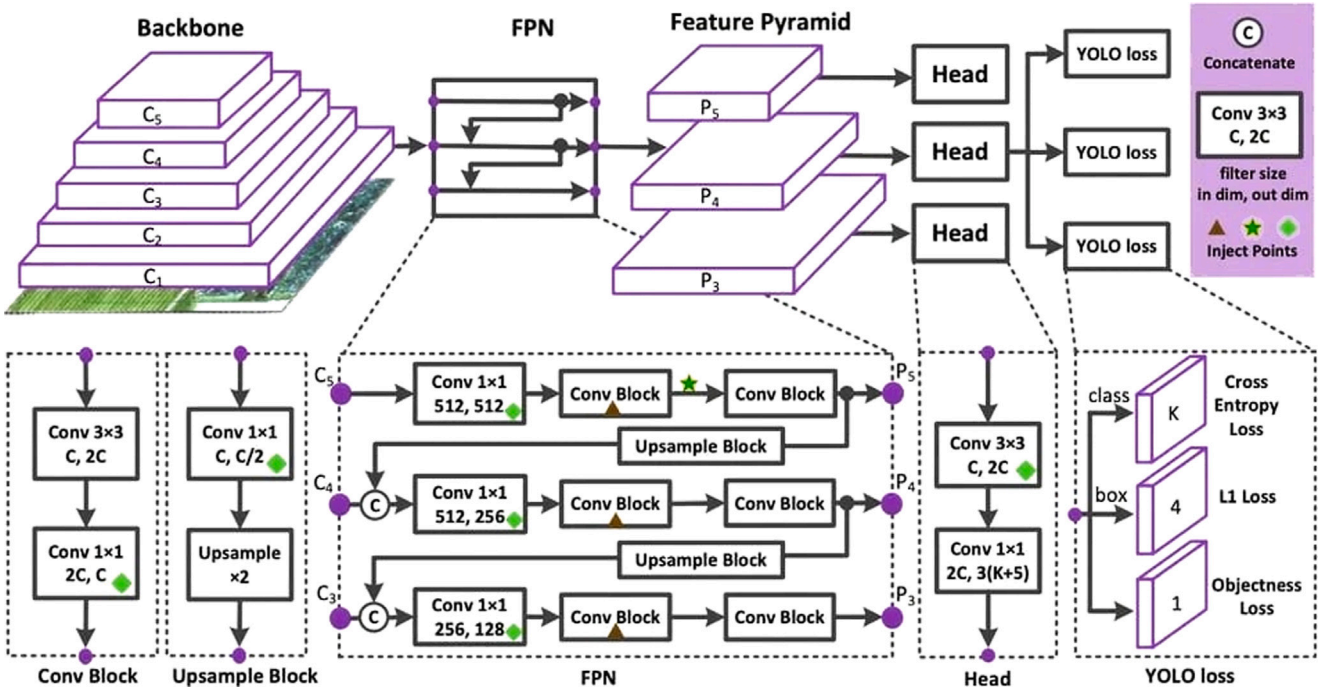


Fig. 3. The architecture of Yolov8 model.

5.2. YOLOv8-based detection

The core of the detection methodology relies on the YOLOv8 model, which is well-suited for the real-time detection and classification of skin lesions. YOLOv8 builds upon previous YOLO versions by incorporating a more refined architecture, including improved backbone networks, better feature aggregation, and sophisticated loss functions, making it highly effective for complex detection tasks such as skin lesion analysis (Bochkovskiy et al., 2020) (see Fig. 3).

YOLOv8 utilizes a CSPDarknet53 backbone, designed to extract rich feature representations from the input images across multiple scales. CSPDarknet53 partitions the feature map into two parts, allowing more gradient flow through the network, which improves feature extraction and model efficiency (Wang et al., 2020). The backbone processes the images to generate feature maps that capture both fine-grained details and high-level semantic information, which are crucial for accurately detecting skin lesions. These feature maps are then passed through the neck of the network, typically using a Path Aggregation Network (PANet), which combines features from different scales to enhance the model's ability to detect objects of varying sizes, a common challenge in the Fredrick Skin Disease Dataset where lesions can range from small acne spots to larger areas of dermatitis.

The model is trained using a composite loss function that balances localization, confidence, and classification losses. The localization loss, $\mathcal{L}_{\text{loc}}$, measures the accuracy of the predicted bounding boxes against the ground truth and is computed as:

$$\mathcal{L}_{\text{loc}} = \sum_{i=0}^{S^2} \sum_{j=0}^{B} \mathbb{1}_{ij}^{\text{obj}} \left[(x_i - \hat{x}_i)^2 + (y_i - \hat{y}_i)^2 + (w_i - \hat{w}_i)^2 + (h_i - \hat{h}_i)^2 \right]$$

The confidence loss, $\mathcal{L}_{\text{conf}}$, accounts for the discrepancy between the predicted confidence scores and the actual IoU (Intersection over Union) between the predicted and ground truth boxes:

$$\mathcal{L}_{\text{conf}} = \sum_{i=0}^{S^2} \sum_{j=0}^{B} \mathbb{1}_{ij}^{\text{obj}} (\hat{C}_i - C_i)^2 + \lambda_{\text{noobj}} \sum_{i=0}^{S^2} \sum_{j=0}^{B} \mathbb{1}_{ij}^{\text{noobj}} (\hat{C}_i - C_i)^2$$

Finally, the classification loss, $\mathcal{L}_{\text{cls}}$, is defined as:

$$\mathcal{L}_{\text{cls}} = \sum_{i=0}^{S^2} \mathbb{1}_{i}^{\text{obj}} \sum_{c \in \text{classes}} (P_i(c) - \hat{P}_i(c))^2$$

These loss components are combined into a total loss function $\mathcal{L}$ as:

$$\mathcal{L} = \lambda_{\text{loc}} \mathcal{L}_{\text{loc}} + \lambda_{\text{conf}} \mathcal{L}_{\text{conf}} + \lambda_{\text{cls}} \mathcal{L}_{\text{cls}}$$

where λ_{loc}, λ_{conf}, and λ_{cls} are hyperparameters that control the contribution of each loss component to the overall training objective (Bochkovskiy et al., 2020).

A significant challenge with the Fredrick Skin Disease Dataset is the imbalance between different classes of skin lesions. For example, there may be a higher number of images depicting common conditions like Acne compared to rarer conditions like Leprosy. To address this, the model integrates a focal loss component into the classification loss. Focal loss helps by focusing on harder-to-classify examples, effectively reducing the impact of class imbalance:

$$\mathcal{L}_{\text{focal}} = -\alpha (1 - p_t)^{\gamma} \log(p_t)$$

where p_t is the predicted probability for the true class, α is a balancing factor, and γ is the focusing parameter (Lin et al., 2017). This modification ensures that the model does not become biased towards the more prevalent classes in the dataset.

Detecting small lesions like Acne or Keratosis Pilaris poses a unique challenge due to their size and subtle appearance. YOLOv8 addresses this challenge through its multi-scale detection approach, which uses feature maps of different resolutions to ensure that smaller lesions are adequately detected. This approach is particularly beneficial for the Fredrick Skin Disease Dataset, where lesions can vary significantly in size. The anchor-free detection heads in YOLOv8 further enhance the model's ability to adapt to the varying shapes and sizes of lesions, improving detection accuracy across all classes.

The model was trained using the *AdamW* optimizer, which was selected for its ability to handle large datasets with high-dimensional features and its efficient weight decay regularization. The learning rate was set to 0.001 to ensure stable convergence as training progressed. A momentum value of 0.9 was used to accelerate gradient vectors in the correct direction, smoothing out the training process and preventing oscillations. To prevent overfitting and enhance generalization, weight decay was applied with a value of 0.0005, helping to control the complexity of the model by penalizing large weights. The focal loss was configured with $\alpha = 0.25$ and $\gamma = 2.0$ to emphasize hard-to-classify

samples and mitigate class imbalance. The model was trained with a batch size of 16 and an image input size of 640 × 640 pixels. This batch size was found to provide a good trade-off between model convergence speed and memory constraints, while the image size allowed the model to capture sufficient detail from the skin lesion images.

During the inference phase, the YOLOv8 model processes each input image to generate bounding box predictions for the detected skin lesions. These predictions are refined using Non-Maximum Suppression (NMS) to eliminate redundant boxes that overlap significantly. The Intersection over Union (IoU) is used as the criterion for NMS:

$$\text{IoU}(A, B) = \frac{A \cap B}{A \cup B}$$

where A and B represent the areas of two bounding boxes. NMS ensures that only the most accurate bounding boxes are retained, providing a clean and precise output (Ren et al., 2015).

5.3. Explainability with saliency maps

Explainability is crucial in medical applications, where trust and interpretability are paramount. In this study, we employ saliency maps to visually explain the decision-making process of the YOLOv8-based skin lesion detection model. Saliency maps reveal which regions of an input image most influence the model's predictions, providing insights into the model's focus during classification and localization tasks (Lopez et al., 2017; Codella et al., 2017).

Saliency maps are derived by calculating the gradient of the model's output with respect to the input image. For a given class c, the saliency map S_c is defined as:

$$S_c = \frac{\partial f_c(I)}{\partial I}$$

where $f_c(I)$ is the model's output score for class c, given an input image I. The gradient $\frac{\partial f_c(I)}{\partial I}$ captures the sensitivity of the model's output to each pixel in the image. Pixels with higher gradient values are those that, when altered, would most significantly impact the model's prediction for class c. This gradient information is visualized as a heatmap over the original image, where the intensity of the color corresponds to the importance of each pixel in the model's decision-making process (Acharya et al., 2019).

Given the complexity of skin lesion detection, where lesions can vary greatly in size, shape, and appearance, we produce composite saliency maps that combine the gradients from both the classification and localization outputs. This approach provides a more comprehensive understanding of the model's focus, ensuring that it is attending to clinically relevant features such as lesion borders, textures, and color variations, rather than being misled by irrelevant background patterns or artifacts (Lopez et al., 2017). For a bounding box B, the saliency map S_B is calculated as:

$$S_B = \frac{\partial \mathcal{L}_{\text{loc}}}{\partial I}$$

where $\mathcal{L}_{\text{loc}}$ is the localization loss associated with the bounding box B. This gradient indicates which parts of the image the model considers most important when determining the position and size of the bounding box, helping to understand how the model localizes lesions within an image (Khan et al., 2019; Sharif et al., 2018). Moreover, the integration of saliency maps with YOLOv8's anchor-free detection heads is a novel contribution of this work. Anchor-free detection heads allow for more flexible bounding box predictions, which are better suited for irregularly shaped lesions. By combining this flexibility with saliency map visualizations, we provide a deeper understanding of how the model handles varying lesion shapes and sizes, particularly in challenging cases where lesions are small or have ambiguous borders.

The use of saliency maps in this study not only enhances the interpretability of the YOLOv8 model but also contributes to the broader field of explainable AI in medical imaging. By providing visual explanations of the model's predictions, we can ensure that the model's decisions are transparent and based on relevant medical features. This transparency is crucial for gaining the trust of healthcare professionals and for further refining the model to improve its accuracy and reliability (Lopez et al., 2017; Codella et al., 2017).

6. Results and evaluation

For evaluation purposes, the dataset was split into three distinct sets: training, validation, and testing. The training set is 85% of the augmented dataset, used to optimize the model's weights and learn the underlying patterns of the skin lesions. A validation set, accounting for 10%, was utilized to tune hyperparameters and monitor the model's performance during training, ensuring that the model did not overfit. Finally, the remaining 5% was reserved as the test set, used exclusively to evaluate the model's generalization performance on unseen data. Metrics such as accuracy, precision, recall, and F1-score were computed on the test set to assess the effectiveness of the model in detecting various skin lesion types.

6.1. Performance evaluation

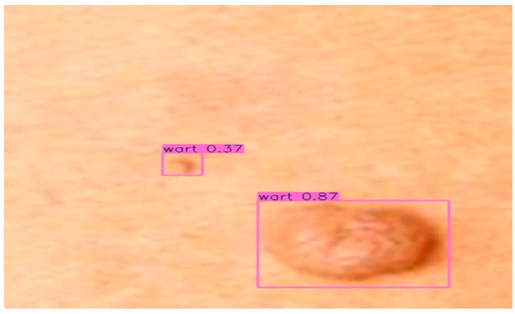
In this section, we evaluate the system's performance across multiple metrics to provide a comprehensive understanding of the system's strengths and limitations.

6.1.1. Detection and explainability analysis

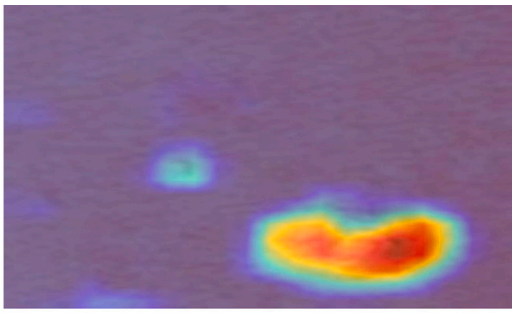
In this section, we present and discuss the results obtained from the proposed skin lesion detection system, with a particular focus on the detection accuracy and the explainability provided by saliency maps. The results are depicted in Fig. 4, which showcase the system's ability to detect and classify different types of skin lesions, such as warts, leprosy, and acne, and highlight the regions of the image that most influenced the model's decisions.

The detection results indicate that our model effectively identifies various skin lesions with high confidence, demonstrating its robustness and reliability across a range of dermatological conditions. The figures illustrate the model's consistency in accurately detecting multiple instances of warts, leprosy, and acne, with confidence scores ranging from 0.29 to 0.89. Notably, the model achieves high confidence in the detection of more prominent lesions, such as larger warts and multiple leprosy patches, which is critical for ensuring reliable diagnosis in clinical settings. On the other hand, The ability to detect smaller and less prominent lesions, as seen in the wart detection image where a smaller wart is identified with a confidence score of 0.37, underscores the model's sensitivity and precision. This performance is a direct result of the advanced data augmentation techniques employed during training, which have significantly enhanced the model's generalization capabilities. By simulating the variability commonly encountered in clinical environments, our model is better equipped to recognize a wide spectrum of lesion types under diverse imaging conditions. Moreover, the consistent high confidence scores in detecting multiple lesions within a single image, particularly in the leprosy detection case, demonstrate the model's robustness in handling complex and overlapping lesions. This ability is crucial for dermatological applications, where multiple lesions often coexist and need to be accurately identified and differentiated.

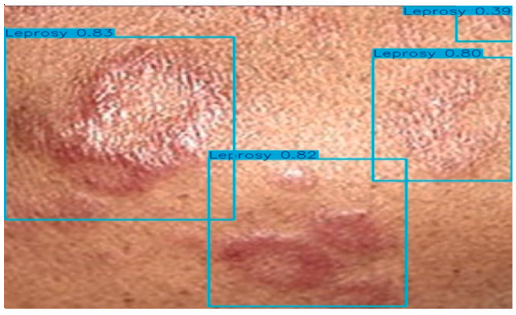
The integration of saliency maps into our framework provides a powerful tool for understanding and validating the model's predictions. These maps visually highlight the regions of the image that the model focuses on when making its decisions, offering transparency and aligning the model's focus with clinically relevant features. In the saliency maps corresponding to the detected lesions, the model's attention is predominantly concentrated on the areas of interest—such as the lesions themselves—while disregarding the surrounding skin. This selective focus on the lesion areas, as indicated by the warmer colors in the saliency maps, confirms that the model's predictions



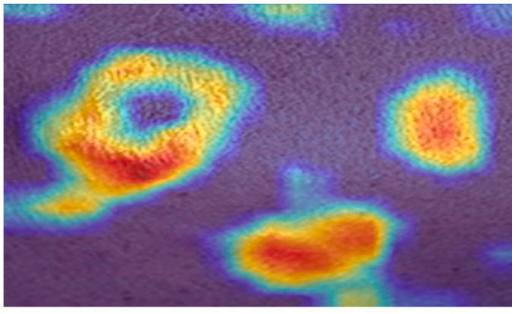
(a) Detection of Warts with confidence scores of 0.37 and 0.87. The model effectively identifies multiple wart lesions with varying levels of confidence.



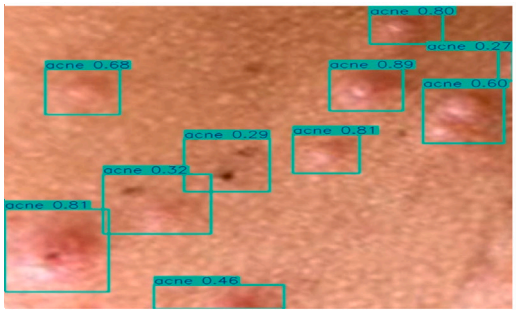
(b) Saliency Map for Wart Detection. The model's attention is focused on the lesion areas, validating the relevance of its predictions.



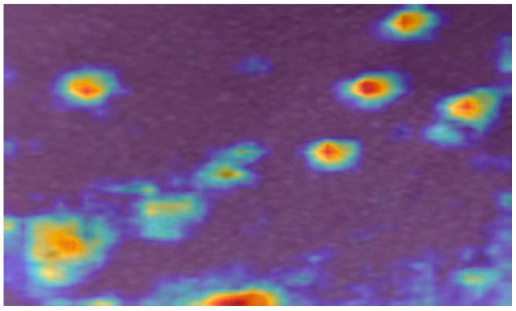
(c) Detection of Leprosy with confidence scores ranging from 0.39 to 0.83. The model demonstrates robustness in detecting multiple and overlapping lesions.



(d) Saliency Map for Leprosy Detection. The model accurately focuses on the distinct lesion areas, ensuring reliable detection.



(e) Detection of Acne with multiple lesions identified, showing confidence scores ranging from 0.27 to 0.89. The model successfully detects both prominent and subtle lesions.



(f) Saliency Map for Acne Detection. The model's attention is effectively concentrated on the acne lesions, confirming the accuracy of its predictions.

Fig. 4. Detection results and corresponding saliency maps for various skin lesions. The left column shows the model's detection outputs with confidence scores, while the right column presents the corresponding saliency maps, highlighting the regions most influential in the model's decision-making process.

are based on clinically significant features like texture, shape, and color variations that are characteristic of different skin conditions. For instance, in the saliency map for wart detection, the model clearly prioritizes the wart itself, validating that the decision-making process is grounded in relevant clinical observations. Similarly, in the leprosy and acne cases, the model's attention is accurately centered on the lesions, further reinforcing the reliability of its predictions. This clear alignment between the model's focus and the actual lesion areas builds trust in the system and ensures that its outputs are interpretable and clinically useful. The use of saliency maps not only enhances the transparency of the model but also serves as a critical validation tool, providing clinicians with visual explanations that support the model's predictions. This explainability is essential for integrating AI systems into clinical practice, as it allows healthcare professionals to understand and trust the model's outputs, leading to more informed and confident decision-making.

6.1.2. Detection performance evaluation

The performance of the proposed skin lesion detection system is evaluated using several key metrics: Precision, Recall, mean Average Precision at 50% IoU (mAP50), and F1-Score. Fig. 5 presents the results across these evaluation metrics, offering a comprehensive overview of the system's effectiveness in accurately identifying and classifying skin lesions. The system achieves a Precision of 90.4%, indicating that the majority of detected lesions are correctly classified, with a low rate of false positives. This high precision is particularly important in medical diagnostics, where minimizing false positives reduces the risk of unnecessary patient anxiety and additional medical procedures. The Recall is 85.6%, reflecting the system's ability to correctly identify

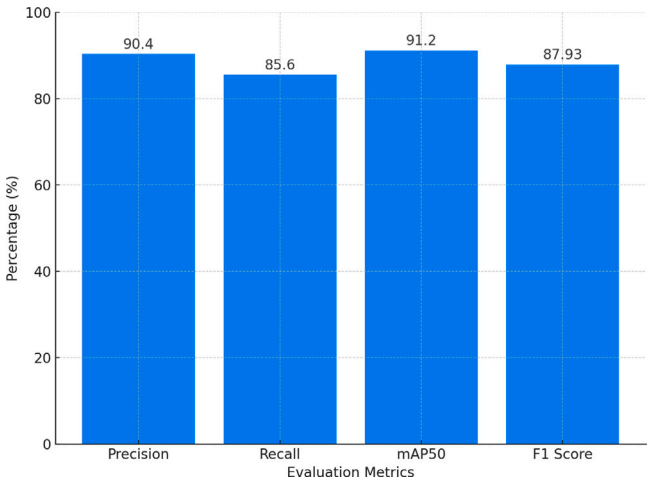


Fig. 5. The system performance evaluation.

relevant lesions. This result demonstrates that the model is effective in minimizing false negatives, thus ensuring that clinically significant lesions are not overlooked. While slightly lower than the precision, this recall value suggests that the system maintains a conservative detection strategy, favoring accurate detections over excessive sensitivity. The system also achieves a mAP50 of 91.2%, highlighting its superior capability in object detection tasks where both classification and precise localization are required. This high mAP50 score indicates that the model effectively aligns predicted bounding boxes with ground truth annotations, which is crucial for dermatological applications where lesion boundaries must be accurately captured. Finally, the F1-Score is 87.93%, representing the harmonic mean of precision and recall. This balanced metric reflects the model's ability to provide consistent performance across both detection and classification tasks. A high F1-Score underscores the reliability and clinical suitability of the proposed framework, demonstrating that it effectively balances the trade-off between missing important cases (recall) and avoiding false alarms (precision).

Overall, these results confirm the proposed system's robustness and potential for deployment in real-world clinical environments, where accurate, reliable, and fast skin lesion detection is essential for effective diagnosis and patient care.

6.1.3. Model performance analysis

The model's performance across various confidence thresholds and lesion classes is illustrated in Fig. 6. The F1-confidence curve (Fig. 6(a)) reveals the model's ability to achieve a balanced trade-off between precision and recall, with a peak F1 score of 0.83 at a confidence threshold of 0.54 for all classes combined. This score indicates that the model performs consistently across most lesion types. The precision–recall curve (Fig. 6(b)) provides further detail on the model's trade-offs between precision and recall across different classes. Notably, Keratosis Pilaris demonstrates near-perfect precision and recall, which is indicative of the model's robustness in detecting this class. On the other hand, Acne shows a decline in precision as recall increases, suggesting that the model struggles with false positives when detecting this class. The overall mAP@0.5 value of 0.858 confirms that the model performs well overall. Fig. 6(c) presents the precision-confidence curve, which shows how the model's precision remains consistently high across most confidence levels, especially for Keratosis Pilaris and Psoriasis. However, Acne poses a challenge, as its precision only improves significantly at higher confidence levels. Finally, the recall-confidence curve (Fig. 6(d)) provides insights into the model's sensitivity at various confidence thresholds. For most classes, recall remains relatively high, even at lower confidence levels, particularly for Psoriasis and Keratosis Pilaris. However, Acne and Leprosy demonstrate a drop in recall as confidence increases, indicating that the model may miss more true positives for these lesions when operating at higher confidence thresholds.

6.1.4. Training progress and overfitting prevention

Fig. 7 displays the training and validation loss curves over the course of model training. The gradual reduction in box loss, classification loss, and DFL loss indicates that the model is learning effectively without overfitting. Interestingly, the validation losses mirror this trend, suggesting good generalization on unseen data. The precision and recall metrics improve as the model progresses, converging around 50 epochs, where the mAP50 metric reaches approximately 0.85. These trends suggest that the data augmentation strategies and loss functions employed were effective in mitigating class imbalance and improving model robustness.

6.2. Ablation study

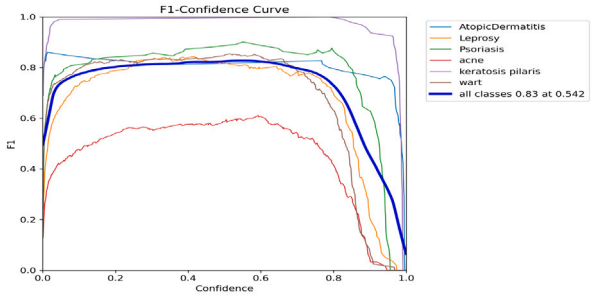
In this section, we delve deeper into specific aspects of the system's performance and design choices. The next subsections are structured to provide a detailed analysis of these elements: Model Performance Under Different Augmentation Strategies, Impact of Loss Functions, Feature Importance and Sensitivity Analysis.

6.2.1. The effect of data augmentation

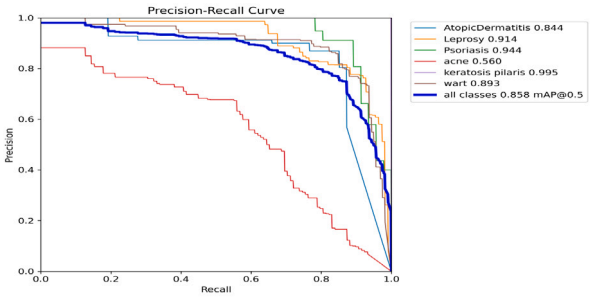
In this section, we study the effect of different data augmentation techniques on the system's performance. Fig. 8 illustrates the impact of these techniques on the F1-Score, revealing that each augmentation strategy contributes positively to the model's ability to detect and classify skin lesions accurately. Specifically, the model trained with no augmentation serves as a baseline, achieving an F1-Score of 79.7. This relatively lower score highlights the challenges associated with training a model on a limited and potentially biased dataset without variations introduced during training. In contrast, applying augmentation techniques such as flipping, noise addition, exposure adjustment, brightness adjustment, and rotation resulted in significant performance improvements. Notably, the F1-Score peaked at 88.0 when brightness adjustment was applied. The improvement in performance with brightness and exposure adjustments emphasizes how these augmentations help the model become more robust to real-world variations in imaging conditions. Such variations are common in clinical practice due to differences in lighting and imaging devices. The ability to generalize across lighting scenarios is essential for accurate lesion detection, as skin images are often captured in varied environments. Additionally, noise addition and rotation demonstrated notable improvements, with F1-Scores reaching 85.8 and 80.2, respectively. These techniques enhance the model's resilience against image imperfections and diverse orientations, which are typical in clinical and telemedicine datasets. Flipping also provided a moderate boost, yielding an F1-Score of 82.7, and reflects the model's improved ability to generalize to mirrored or rotated lesion appearances. Overall, the analysis clearly demonstrates that data augmentation is not merely an optional enhancement, but a critical step in developing a high-performing and generalizable skin lesion detection model. By systematically improving the model's robustness to visual variability and common distortions, augmentation ensures the system remains effective across diverse clinical imaging conditions.

6.2.2. Robustness to class imbalance

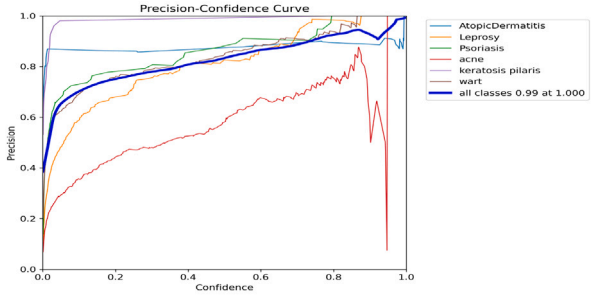
Fig. 10 presents a detailed analysis of the class-wise performance of the proposed YOLOv8n-based detection framework, evaluated in terms of F1-Score and mAP@50 alongside the number of training instances for each lesion class. The dataset exhibits significant class imbalance, a common challenge in dermatological image analysis where rare skin conditions are often underrepresented. Notably, the model achieves exceptionally high performance on both majority and minority classes.



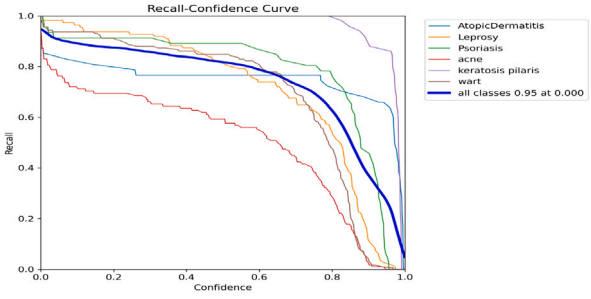
(a) F1-Confidence Curve: Displays the relationship between confidence thresholds and the F1 score for each skin lesion class.



(b) Precision-Recall Curve: Highlights the trade-off between precision and recall for different lesion types.



(c) Precision-Confidence Curve: Shows how precision varies across confidence levels, emphasizing the model's ability to maintain high precision.



(d) Recall-Confidence Curve: Demonstrates how the model's recall changes with increasing confidence thresholds, providing insights into its sensitivity for different classes.

Fig. 6. Evaluation curves for model performance across skin lesion classes.

For instance, Atopic Dermatitis, the least represented class with only 74 training instances, records an F1-Score of 100% and mAP@50 of 99%. Similarly, Psoriasis and Keratosis Pilaris, with 261 and 137 instances respectively, also achieve near-perfect scores—indicating that the model generalizes well even under data-scarce conditions. These results strongly highlight the effectiveness of our augmentation pipeline and training strategies, particularly the use of focal loss to address class imbalance and improve minority class representation during learning. While Acne is the most abundant class (1208 instances), it exhibits comparatively lower performance (F1-Score of 66%, mAP@50 of 65%). This result may initially seem counterintuitive but can be attributed to the higher intra-class variation and the complex visual patterns of acne lesions, which include a mix of papules, pustules, and nodules across different body areas. These features make accurate and consistent detection more challenging. Similarly, Leprosy, with 643 instances, also demonstrates moderate performance, potentially due to its overlapping visual characteristics with other classes and high inter-patient variation. In contrast, Wart, a well-represented class with 807 instances, achieves robust scores (F1-Score of 94%, mAP@50 of 96%), showing that the model can capitalize on sufficient data volume when lesions exhibit consistent visual patterns.

Overall, the framework demonstrates strong resilience to class imbalance, maintaining high detection accuracy across both over- and under-represented categories. This robustness is achieved through a combination of data-level strategies (balanced augmentation, brightness/contrast transformations) and model-level techniques (focal loss, anchor-free detection). These findings affirm the model's practical value for real-world clinical use, where imbalanced datasets are often unavoidable.

6.2.3. Cross-dataset evaluation and generalization analysis

To assess the robustness and generalizability of the proposed YOLOv8n-based framework, we conducted experiments on two distinct public datasets: the Fredrick Skin Disease Dataset (Gk, 2024) and the HAM10000 Skin Diseases Detection dataset (Universe, 2024). Fig. 9 presents a visual comparison of the evaluation metrics across these two datasets. The Fredrick Skin Disease Dataset, used as the primary benchmark in this study, is a curated and augmented dataset specifically designed for academic research in medical image analysis. It offers high-resolution lesion images with consistent labeling and class balance, making it well-suited for training and evaluating deep learning models under structured conditions. On this dataset, the proposed model achieved a precision of 90.4%, recall of 85.6%, F1-score of 87.9%, and mAP@50 of 91.2%, indicating strong performance in both detecting and classifying skin lesions with accurate localization. To validate the framework's ability to generalize beyond this controlled setup, we further evaluated it on the HAM10000 Skin Diseases Detection dataset from Roboflow Universe. This dataset is more diverse in terms of image sources, resolution, and annotation style, reflecting real-world heterogeneity. Unlike the Fredrick dataset, it lacks uniform preprocessing and includes a wider range of imaging artifacts and visual noise. When tested on this dataset, the proposed model attained a precision of 74.1%, recall of 78.2%, F1-score of 76.0%, mAP@50 of 82.4%, and mAP@50–95 of 54.0%. Although these results are lower compared to those on the primary dataset, they demonstrate the model's resilience to domain shifts and its ability to maintain solid detection accuracy under more varied and uncontrolled conditions. The performance differences highlight the challenges introduced by real-world deployment environments, including inconsistent lighting, focus, skin tones, and annotation quality. Despite these challenges, the model preserved robust detection capability, illustrating the architectural strengths of the YOLOv8n-based system, including its anchor-free detection, decoupled heads, and multi-scale feature fusion. These findings reinforce the proposed framework's suitability for clinical use cases, particularly where training and testing distributions often diverge.

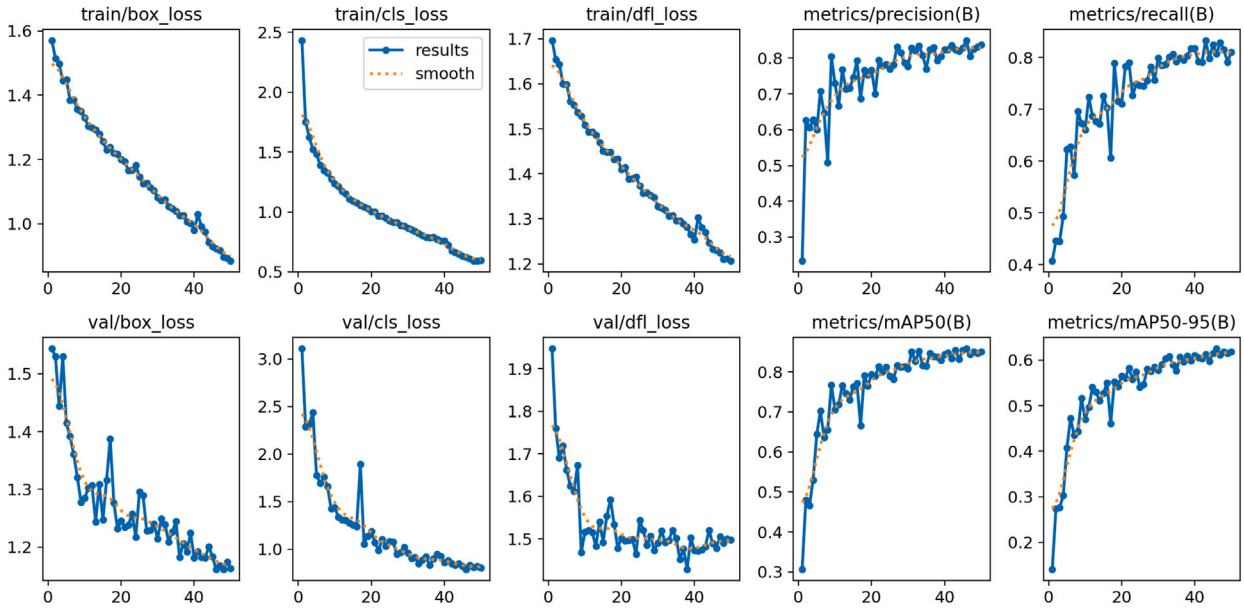


Fig. 7. The training curves.

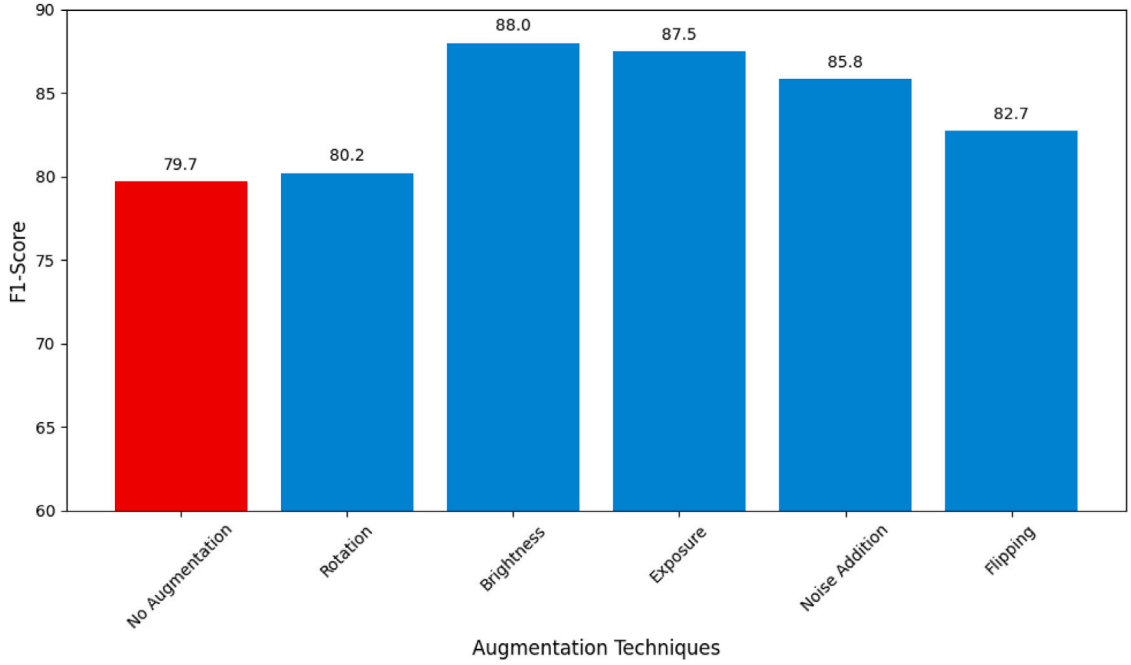


Fig. 8. Impact of different data augmentation techniques on the system performance.

6.3. Comparative evaluation

6.3.1. Comparison of detection performance

Fig. 11 presents a comprehensive evaluation of the proposed YOLOv8n-based skin lesion detection framework against five alternative models: YOLOv9t, YOLOv10n, Vision Transformer (ViT), Faster R-CNN, and RetinaNet. These models represent a variety of detection paradigms including anchor-based, anchor-free, single-stage, two-stage, and attention-based methods, providing a broad view of their respective capabilities in the context of skin lesion detection. Each model was evaluated using four key performance metrics: Precision, Recall, F1-Score, and mean Average Precision at 50% IoU (mAP50).

Within the YOLO family, YOLOv9t employs a transformer-based detection head, introducing self-attention mechanisms into the architecture. This allows the network to better capture long-range dependencies and contextual cues, which is beneficial for distinguishing lesions from complex backgrounds and for detecting clear and well-defined lesion boundaries. However, this attention-driven design introduces higher detection confidence thresholds and suppresses lower-confidence predictions, which leads to high precision (91.7%) but lower recall (80.5%). As a result, the model yields an F1-Score of 85.2% and mAP50 of 89.7%, indicating reduced sensitivity to subtle or faint lesions, which can be critical in medical applications. YOLOv10n, in contrast, prioritizes speed and resource efficiency. It achieves this through aggressive pruning and architectural simplification, resulting in a lightweight model optimized for edge device deployment.

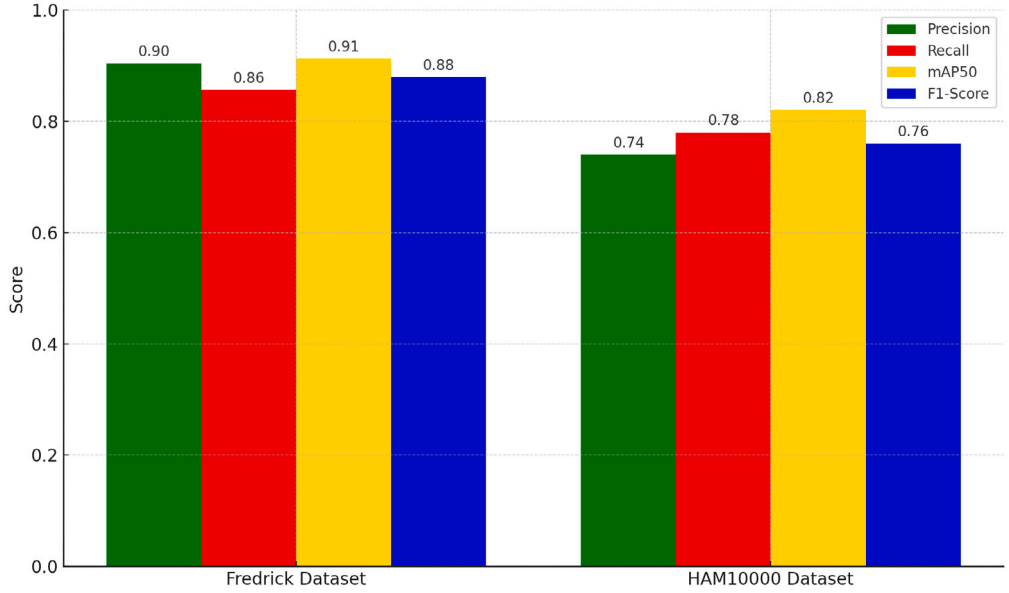


Fig. 9. Impact of different data augmentation techniques on the system performance.

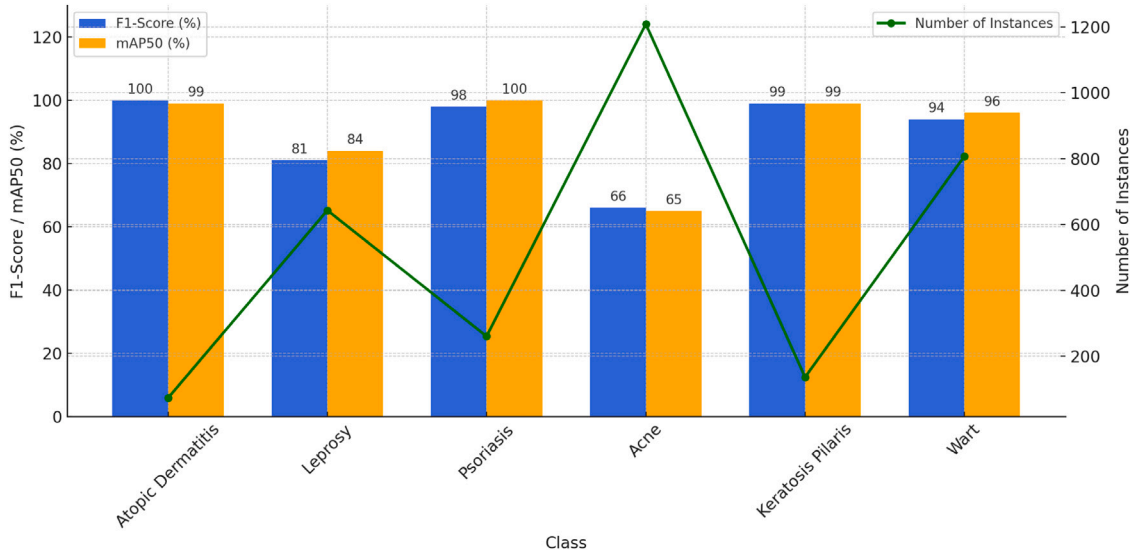


Fig. 10. Impact of different data augmentation techniques on the system performance.

This reduction in model capacity increases recall (84.8%) by lowering detection thresholds and increasing sensitivity to weak signals, but simultaneously reduces precision (82.2%) due to a higher number of false positives. This trade-off leads to a lower F1-Score of 78.9% and mAP50 of 87.5%, reflecting a model that is faster but less reliable for nuanced lesion detection tasks.

The Proposed YOLOv8n model achieves the best balance between detection accuracy and efficiency. It integrates decoupled heads for classification and localization, anchor-free object detection, and a CSP-Darknet backbone with PANet for multi-scale feature aggregation. This architectural design allows the network to efficiently process input at different scales and maintain precise localization while avoiding the drawbacks of anchor-based detection. The result is a well-rounded model with a precision of 90.4%, recall of 85.6%, F1-Score of 87.9%, and mAP50 of 91.2%, making it highly suitable for real-time and reliable clinical decision support.

The Vision Transformer (ViT)-based detector represents a fundamentally different design philosophy. It replaces convolution with self-attention mechanisms across its layers, enabling global feature interactions and robust generalization across lesion types and appearances. ViT achieved competitive precision (86.2%) and recall (85.1%), leading to an F1-Score of 85.6% and mAP50 of 88.5%. However, the lack of inductive biases (such as locality and translation equivariance) and high computational overhead make ViT less practical for real-time or low-latency clinical deployments, despite its strong recognition capability. On other hand, Faster R-CNN follows a two-stage detection paradigm, where region proposals are first generated and then refined with a separate classification and regression head. This approach generally provides high accuracy in generic object detection, but it is computationally expensive and struggles with small or subtle objects like skin lesions. In this study, it achieved a precision of 59.6%, recall of 71.0%, F1-Score of 64.8%, and mAP50 of 55.0%, reflecting significant challenges in both localization and classification of lesions. Similarly, RetinaNet is a single-stage detector that introduces focal

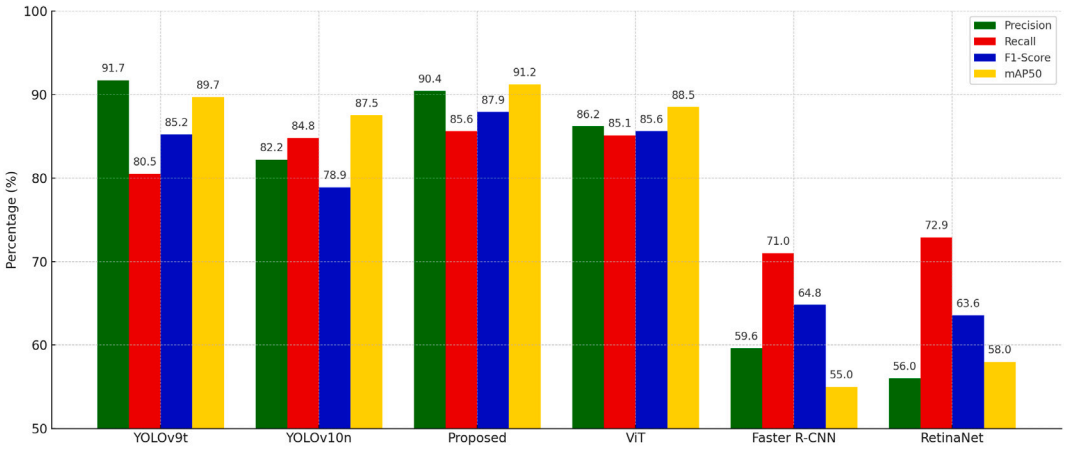


Fig. 11. Comparison of Precision, Recall, F1-Score, and mAP50 between YOLOv9t, YOLOv10n, the proposed model, and a Transformer-based detector, demonstrating the superior balance of performance and efficiency in the proposed approach.

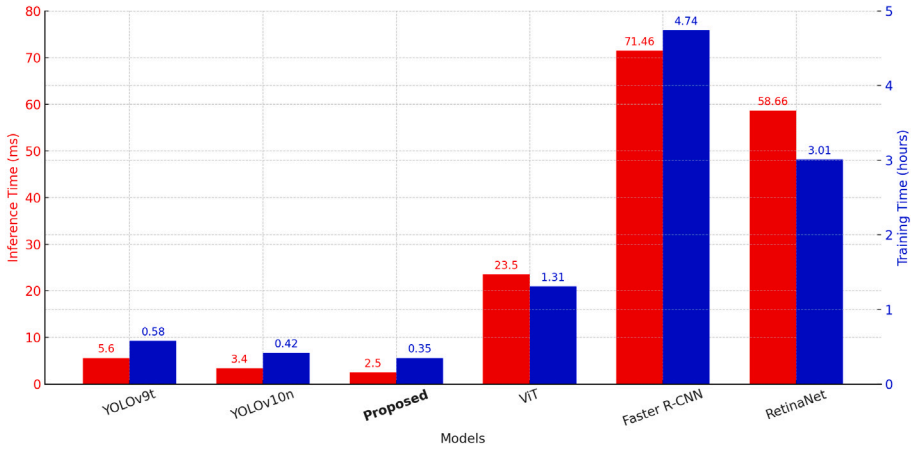


Fig. 12. Comparison of training and inference times for YOLOv9t, YOLOv10n, the proposed model, a Vision Transformer-based detector, RetinaNet, and Faster R-CNN, highlighting the superior computational efficiency of the proposed approach for real-time skin lesion detection.

loss to address class imbalance, making it suitable for datasets where foreground objects are sparse. Although it improves over traditional single-stage methods, RetinaNet still falls short in this application, achieving precision of 56.0%, recall of 72.9%, F1-Score of 63.6%, and mAP50 of 58.0%. This indicates that while it is able to detect many lesions (as seen in recall), its ability to precisely classify and localize them is limited, resulting in many false positives.

In conclusion, the proposed YOLOv8n-based framework clearly outperforms both its YOLO successors and non-YOLO alternatives. Its anchor-free design, decoupled heads, and optimized multi-scale feature processing offer the best compromise between speed, precision, and generalization. This makes it not only technically superior in detection performance but also highly practical for trustworthy and real-time clinical skin lesion detection and analysis.

6.3.2. Comparison of training and inference efficiency

This section presents a detailed analysis of the computational efficiency of the proposed YOLOv8n-based framework in comparison with other state-of-the-art models, focusing on training time and inference latency—two critical metrics that directly influence feasibility for real-time deployment in clinical practice. As shown in Fig. 12, the proposed model demonstrates the most advantageous runtime profile among all evaluated methods, completing training in only 0.35 h and achieving ultra-fast inference with a latency of 2.5 ms per image. These outstanding results are largely attributed to YOLOv8n's streamlined design, which integrates anchor-free detection, decoupled heads, and an efficient CSPDarknet backbone combined with PANet-based multi-scale feature aggregation. These architectural optimizations ensure rapid convergence during training and minimize computational demands during inference. In contrast, YOLOv9t, while delivering strong precision due to its transformer-based attention mechanisms, requires a longer training duration (0.58 h) and higher inference latency (5.6 ms). This added overhead reflects the cost of its enhanced context modeling capabilities. YOLOv10n, optimized for edge devices through aggressive parameter pruning, achieves faster training (0.42 h) and inference (3.4 ms), but this comes at the expense of reduced detection performance, emphasizing the trade-off between speed and accuracy in lightweight models. Beyond the YOLO-based models, the Vision Transformer (ViT)-based detector exhibited the highest computational cost. Although it achieved competitive detection accuracy, its training time of 1.31 h and inference latency of 23.5 ms per image render it impractical for real-time or embedded medical applications. Similarly, RetinaNet and Faster R-CNN, both well-established detection frameworks, demonstrated relatively high computational burdens. RetinaNet completed training in 3.01 h with an inference latency of 58.66 ms, while Faster R-CNN required 4.74 h of training and 71.46 ms for inference per image. These results are expected given their dense prediction mechanisms and multi-stage pipelines, which, although effective in general object detection tasks, are less suited to latency-sensitive medical environments.

Taken together, these findings highlight the exceptional computational efficiency of the proposed YOLOv8n-based framework. It not only achieves superior accuracy and robustness in skin lesion detection

but also maintains the fastest inference speed among all evaluated models. This balance makes it particularly well suited for clinical use cases that demand real-time decision support, such as bedside diagnostic aids, teledermatology platforms, and mobile healthcare applications.

6.4. Limitations and deployment challenges

Despite the strong performance and computational efficiency demonstrated by the proposed YOLOv8n-based framework, several practical limitations and deployment challenges remain, which are important to acknowledge when considering real-world clinical integration.

First, the model was trained using two NVIDIA Tesla T4 GPUs, a configuration typical of modern data centers but not necessarily representative of hardware available in all clinical environments. However, the resulting model achieves an inference latency of just 2.5 ms per image and requires only 0.35 h of training, demonstrating a highly optimized computational footprint. These results suggest that the model can be effectively deployed on mid-tier GPU platforms such as NVIDIA RTX 3060, as well as edge inference platforms including NVIDIA Jetson Xavier, Apple Neural Engine (ANE), or mobile NPUs for point-of-care applications. Nevertheless, adaptation to extremely low-power devices (e.g., microcontrollers or IoT nodes) may still require further model quantization or compression techniques such as pruning or knowledge distillation. Second, while our cross-dataset evaluation confirms the model's ability to generalize across varied skin lesion datasets, we recognize that training data diversity remains a limiting factor. The current datasets do not cover the full spectrum of skin tones, imaging modalities, and rare dermatological conditions encountered globally. As a result, model robustness could be further enhanced through training on multi-institutional, multi-national datasets to minimize potential bias and improve fairness across demographics.

From a deployment perspective, regulatory compliance and clinical validation are critical challenges. Although our framework demonstrates strong explainability and real-time performance, clinical deployment will require rigorous validation and regulatory clearance, such as FDA approval in the United States and CE marking in Europe. To prepare for this, we have initiated collaboration with medical institutions in Japan and are in the process of preparing documentation for Institutional Review Board (IRB) approval. Future work will involve prospective user studies with dermatologists, capturing both qualitative trust assessments and integration feedback, which are essential components for evidence-based certification.

Finally, while the system supports interpretability via saliency map visualization, there remains an opportunity to extend the explainability features using multimodal input (e.g., combining clinical metadata with image analysis), or interactive interfaces that better align with clinicians' decision-making workflows.

7. Conclusions

In this paper, we have proposed a robust and explainable YOLOv8-based framework for the detection and classification of skin lesions, addressing key challenges in medical imaging, particularly within the dermatological domain. Our approach integrates advanced data augmentation techniques to enhance the model's generalization capabilities across a diverse set of clinical conditions. This enables our framework to perform well even in the presence of significant class imbalance, a common challenge in medical datasets. Through comprehensive evaluation, we demonstrated the effectiveness of our framework in achieving high accuracy across various skin lesion classes, including both common and rare conditions such as Atopic Dermatitis, Leprosy, Psoriasis, Acne, Keratosis Pilaris, and Warts. The model's ability to maintain strong performance despite the variability in data availability and lesion complexity highlights its robustness and reliability for potential real-world clinical applications. A key contribution of this work is the integration of saliency maps, which provide visual explanations for the model's predictions. This explainability component is crucial for building trust among clinicians, as it allows them to understand the decision-making process of the AI model and ensures that the model's focus aligns with clinically relevant features. By offering transparency in model predictions, we bridge the gap between high-performance AI models and the interpretability required for their deployment in healthcare settings. Evaluation across multiple public datasets demonstrated that our system consistently outperforms several state-of-the-art models—including YOLOv9t, YOLOv10n, Faster R-CNN, RetinaNet, and a Vision Transformer—in terms of precision, recall, F1-score, mAP@50, and computational efficiency.

Despite the model's strong performance, there are several avenues for future improvement. One key area is the inclusion of additional data from underrepresented populations, such as individuals with darker skin tones, which would help mitigate any potential biases. Moreover, leveraging federated learning could enhance the model's ability to adapt to new data sources without compromising patient privacy, a critical requirement for clinical deployment. Finally, we will pursue regulatory pathways and conduct prospective user studies with dermatologists to evaluate diagnostic trust, usability, and deployment viability in real-world settings.

CRediT authorship contribution statement

Mohammad A. Eita: Writing – review & editing, Supervision, Software, Methodology, Data curation, Conceptualization. **Hamada Rizk:** Writing – original draft, Visualization, Formal analysis.

Declaration of competing interest

The authors declare that they have no conflicts of interest regarding the publication of this paper. They affirm that this manuscript is original.

Data availability

You can find the dataset at: https://universe.roboflow.com/fredrick-gk-wpenz/skin-disease-hwlrn.

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